

FORM TP 2011044

CANDIDATE – PLEASE NOTE!

You must sign below and return this booklet with the Answer Sheet. Failure to do so may result in disqualification.

Signature _____

TEST CODE **01207010**

MAY/JUNE 2011

CARIBBEAN EXAMINATIONS COUNCIL**SECONDARY EDUCATION CERTIFICATE
EXAMINATION****BIOLOGY****Paper 01 – General Proficiency***75 minutes***06 JUNE 2011 (p.m.)****READ THE FOLLOWING INSTRUCTIONS CAREFULLY.**

1. This test consists of 60 items. You will have 75 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

Which of the following diseases is due to a dietary deficiency?

- (A) Malaria
(B) Diabetes
(C) Influenza
(D) Anaemia

Sample Answer

The best answer to this item is "Anaemia", so answer space (D) has been shaded.

5. If you want to change your answer, erase it completely and fill in your new choice.
6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, omit it and go on to the next one. Your score will be the total number of correct answers.
7. Figures are not necessarily drawn to scale.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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01207010/F 2011

Item 1 refers to the following food chain which shows the feeding relationships in a freshwater habitat.

Microscopic alga → Mosquito larva → Small fish → Large fish

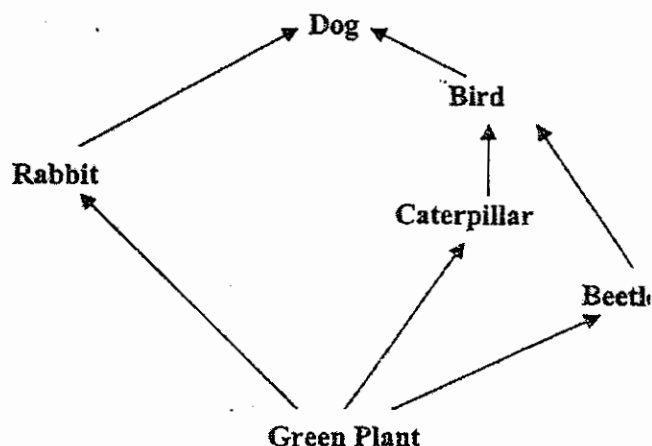
1. The organism to which the LEAST amount of energy is available is the
- (A) small fish
 - (B) large fish
 - (C) mosquito larva
 - (D) microscopic alga

Item 2 refers to the following animals which were classified using the visible characteristics listed below.



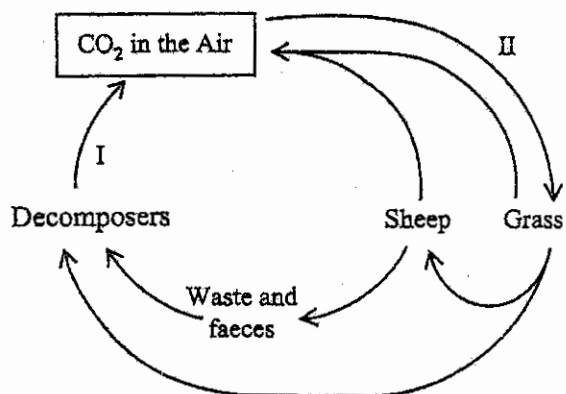
- I. Shape
 - II. Number of wings
 - III. Number of legs
 - IV. Number of eyes
 - V. Number of antennae
2. Which characteristics could be used to divide the animals into two groups?
- (A) I and III only
 - (B) II and V only
 - (C) III and IV only
 - (D) IV and V only

Item 3 refers to the following food web which was observed in a small terrestrial habitat.



3. Which of the following pairs of organisms is at the same trophic level?
- (A) Beetle, caterpillar
 - (B) Dog, beetle
 - (C) Bird, rabbit
 - (D) Dog, rabbit
4. When does respiration occur in plants?
- (A) All the time
 - (B) Only at night
 - (C) When photosynthesis stops
 - (D) Only while photosynthesis is taking place

Item 5 refers to the following diagram.



5. What are the processes taking place at I and II?

	I	II
(A)	Respiration	Combustion
(B)	Photosynthesis	Combustion
(C)	Respiration	Photosynthesis
(D)	Decomposition	Photosynthesis

6. The variation in cellular structure found in any complex organism is due to a process called

- (A) growth
(B) elaboration
(C) metabolism
(D) specialization

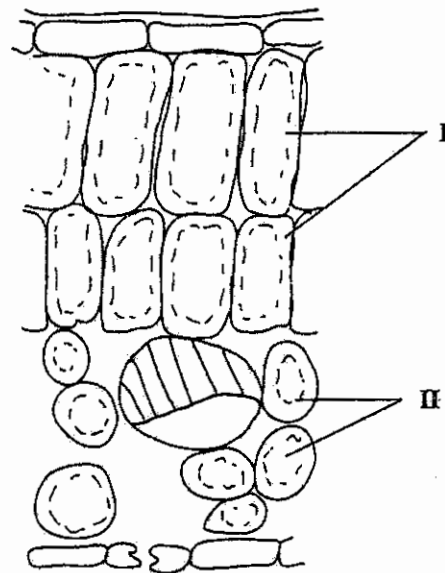
7. Which of the following pairs of function and organelle is NOT correctly matched?

	Function	Organelle
(A)	Osmotic control	Membrane
(B)	Protein synthesis	Chloroplast
(C)	Hereditary materials	Nucleus
(D)	Release of energy	Mitochondrion

8. Energy is released from sugar during

- (A) photosynthesis
(B) transpiration
(C) respiration
(D) nutrition

Item 9 refers to the following diagram which shows a cross-section through a dicotyledonous leaf.

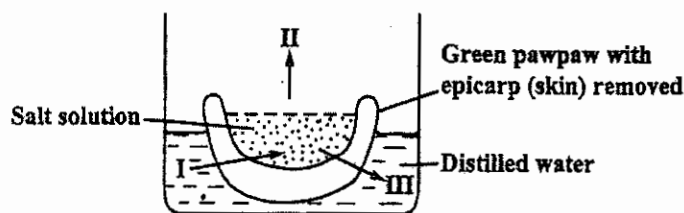


9. Which of the following is NOT true of the cells labelled I and II in the diagram above?

- (A) I makes most of the food.
(B) I has a relatively larger surface area.
(C) II receives less energy from the sun.
(D) II possesses more chloroplasts than I.

10. Plant cells differ from animal cells in their ability to
- (A) control the intake of ions
 - (B) control the outflow of ions
 - (C) resist turgor pressure
 - (D) resist plasmolysis

Item 11 refers to the apparatus below which was set up in a school yard on a sunny day.



11. After 20 minutes, movement of water should have occurred in the direction(s) of

- (A) I only
- (B) I and II only
- (C) I and III only
- (D) I, II and III

12. Which of the following structures found in the leaves of a flowering plant contain chloroplasts?

- I. Epidermis
- II. Mesophyll layer
- III. Guard cells
- IV. Vein

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) II and IV only

13. Which of the following teeth would be MOST suitable for peeling cane or removing tough meat from a bone?

- (A) Incisors
- (B) Canines
- (C) Premolars
- (D) Molars

14. The part of the gut into which bile empties is the

- (A) ileum
- (B) caecum
- (C) stomach
- (D) duodenum

15. West Indian cherries are rich in Vitamin C. Which of the following BEST describe(s) the benefit a boy derives from consuming West Indian cherries (fruit) everyday?

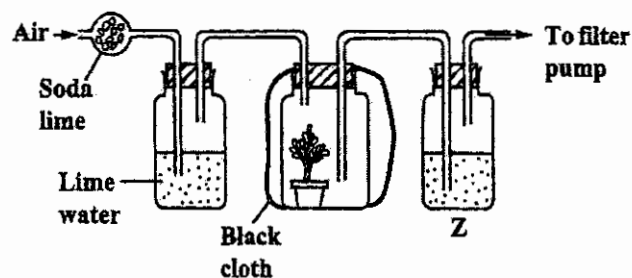
- I. Increase in the absorption of iron
- II. Prevention of bleeding gums
- III. Increase in the absorption of calcium

- (A) I only
- (B) I and II only
- (C) I and III only
- (D) I, II and III

16. Which of the following options is true for aerobic and anaerobic respiration?

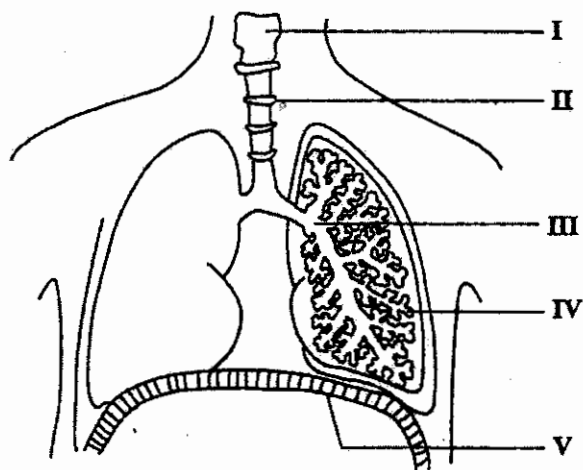
	Aerobic	Anaerobic
(A)	Does not use O_2 .	Uses O_2 .
(B)	Occurs in the cytoplasm.	Occurs in the mitochondria.
(C)	Releases large amounts of energy.	Releases small amounts of energy.
(D)	At least one product is organic.	Products are always inorganic.

set up to test for products of aerobic respiration.



17. The solution in Jar Z should be
- (A) iodine
 - (B) tap water
 - (C) calcium hydroxide (lime water)
 - (D) sodium hydroxide (caustic soda)
18. The energy released in respiration is temporarily captured in
- (A) sucrose
 - (B) glycogen
 - (C) adenosine diphosphate (ADP)
 - (D) adenosine triphosphate (ATP)

Items 19 - 20 refer to the following diagram of the human respiratory system.



19. The site of gaseous exchange is
- (A) I
 - (B) III
 - (C) IV
 - (D) V

20. The volume of air within the lungs is controlled by

- (A) I
- (B) II
- (C) IV
- (D) V

21. As part of the sequence of the heartbeat, the ventricular muscle contracts, the atrio-ventricular valves close, and the arterial valves open. This allows blood to flow from

- (A) veins to the atria
- (B) atria to the ventricles
- (C) ventricles to the veins
- (D) ventricles to the arteries

22. Which of the following organisms has the LEAST need for a transport system?

- (A) Tadpole
- (B) Amoeba
- (C) Seedling
- (D) Honey bee

23. Human red blood cells are normally

- (A) spherical, with a nucleus
- (B) spherical, without a nucleus
- (C) disc-shaped, with a nucleus
- (D) disc-shaped, without a nucleus

24. A whole plant has its roots submerged for a few hours in water coloured with red ink. In which part of its leaves does the ink FIRST show up?

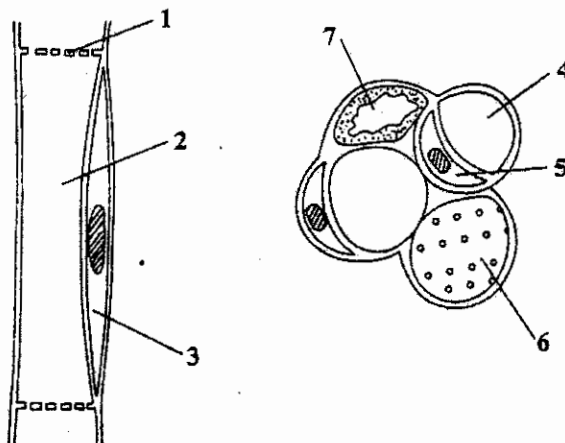
- (A) Vein
- (B) Epidermis
- (C) Guard cells
- (D) Mesophyll layer

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25. Which of the following conditions INCREASES the rate of transpiration?

(A) High humidity
(B) Low temperature
(C) High wind velocity
(D) Low light intensity

Item 26 refers to the following sections of phloem tissues from a plant stem.



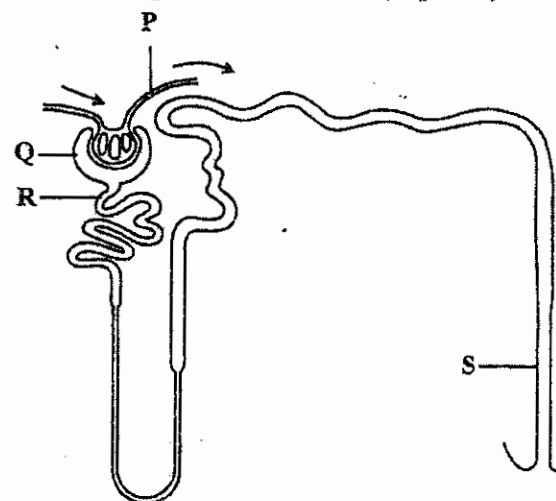
26. Companion cells are represented by

(A) 1 and 7
(B) 2 and 5
(C) 2 and 6
(D) 3 and 5

27. Mineral salts are transported in plants MAINLY through the

(A) cambium
(B) sieve tubes
(C) parenchyma
(D) xylem vessels

Items 28-29 refer to the following simplified drawing of a renal tubule (nephron).



28. The concentration of urea at S is about 60 times its concentration at R because

(A) large quantities of urea are excreted from the body
(B) urea passes into the tubule at various points along its length
(C) most of the fluid in the tubule is reabsorbed before reaching S
(D) at S, urine collects from several smaller collecting ducts

29. One component present in the fluid in P but not in the fluid in Q is

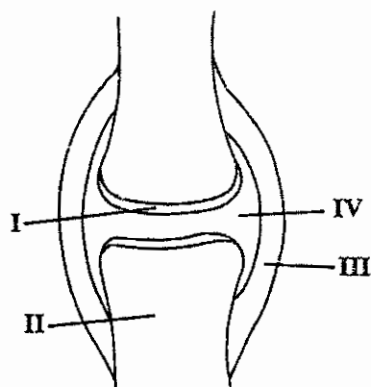
(A) urea
(B) glucose
(C) blood proteins
(D) blood plasma

30. Excretion is important in living organisms because metabolic waste products

(A) cannot be stored
(B) decay very readily
(C) are very soluble substances
(D) can poison living organisms

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Item 31 refers to the following diagram of a section through a joint.



31. Which of the following components reduce friction?

- (A) I and II
- (B) I and IV
- (C) II and III
- (D) II and IV

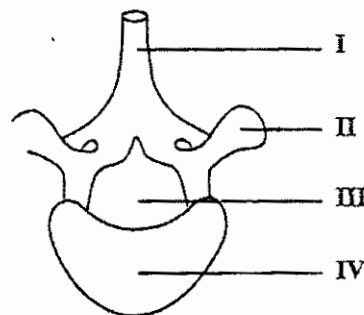
32. The primary function of the neural spine of the lumbar vertebra is for

- (A) attachment of muscles
- (B) movement of bones
- (C) protection of the spinal cord
- (D) support of adjacent vertebrae

33. Which of the following is a function of ligaments?

- (A) Preventing friction
- (B) Holding bones together
- (C) Attaching muscles to bones
- (D) Helping to lubricate joints

Item 34 refers to the following diagram of a thoracic vertebra.



34. Nervous tissue is found at

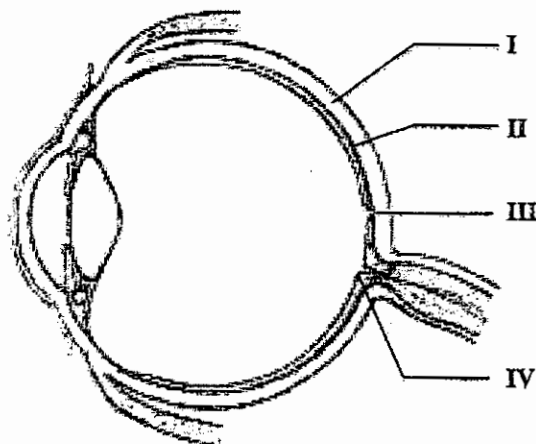
- (A) I
- (B) II
- (C) III
- (D) IV

35. For several days, light reaching some seedlings was limited to one side. The seedlings bent towards the light. Which of the following BEST explain(s) this response?

- I. Uneven growth of the cells at the node of the stem
- II. Uneven growth of the cells at the region behind the tip of the stem
- III. Inhibition of the production of auxins on the unlit side

- (A) I only
- (B) II only
- (C) I and III only
- (D) II and III only

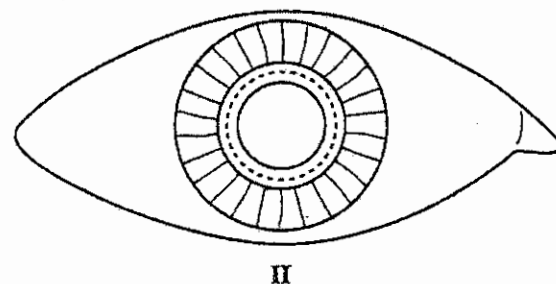
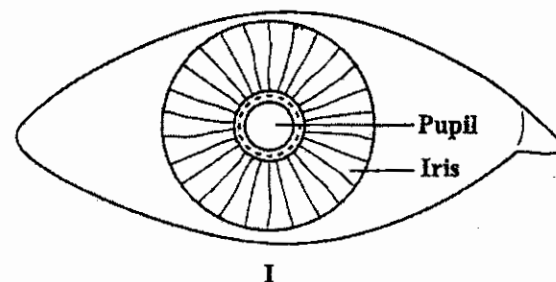
Item 36 refers to the following diagram of a longitudinal section through the human eye.



36. At what point is vision NOT possible?

- (A) I
- (B) II
- (C) III
- (D) IV

Item 37 refers to the following diagrams (I and II) representing the iris in bright and dim light respectively.

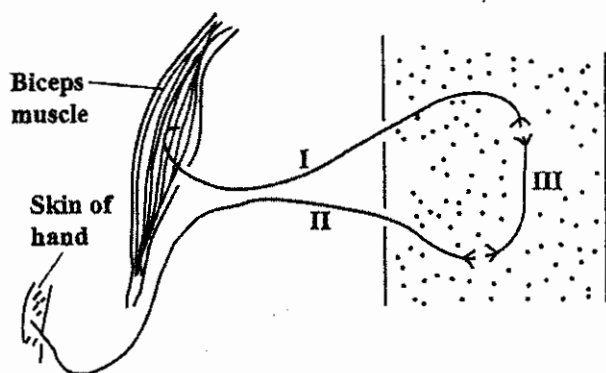


37. What causes the appearance in I and II to be different?

- (A) In bright and dim light the circular and radial muscles both contract.
- (B) In bright and dim light both circular and radial muscles relax.
- (C) In dim light the circular muscles relax and radial muscles contract, and in bright light radial muscles relax and circular muscles contract.
- (D) In bright light circular muscles relax and radial muscles contract, and in dim light radial muscles relax and circular muscles contract.

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Items 38-39 refer to the following diagram.



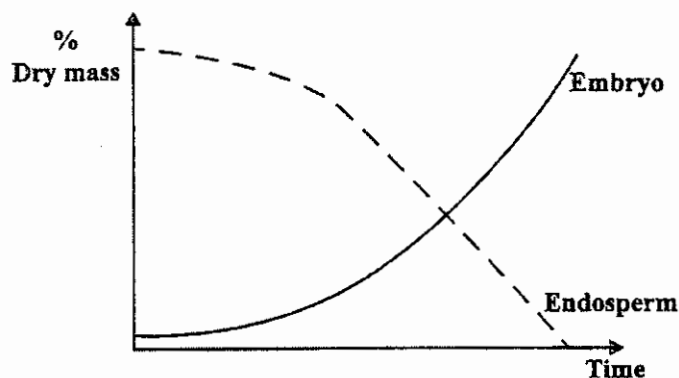
38. When the skin of the hand touches a hot object, the sequence of impulse flow is

- (A) I → II → III
- (B) I → III → II
- (C) II → I → III
- (D) II → III → I

39. The numbered structures are

	I	II	III
(A)	sensory neurone	effector	relay neurone
(B)	motor neurone	sensory neurone	relay neurone
(C)	effector	relay neurone	receptor
(D)	receptor	relay neurone	sensory neurone

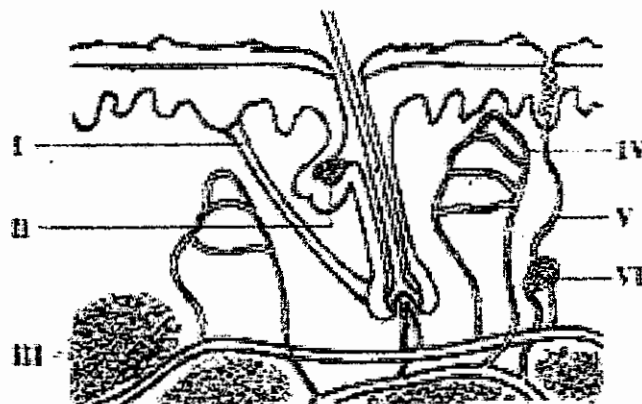
Item 40 refers to the following graph which shows changes in dry mass of embryo and endosperm during germination.



40. From the graph, it is true to say that

- (A) as the embryo grows, the food reserves are gradually depleted
- (B) the embryo and the endosperm decrease in mass at the same rate
- (C) the embryo and the endosperm increase in mass at the same rate
- (D) the embryo begins to grow as the endosperm increases in mass

Item 41 refers to the following diagram of a vertical section through human skin.

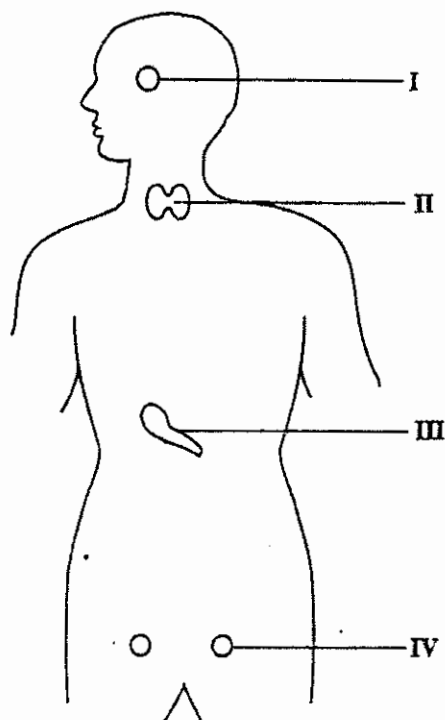


41. Which structure(s) above will be involved in temperature regulation if there is a DECREASE in external temperature?

- (A) I only
- (B) II and V only
- (C) I, III and IV only
- (D) IV, V and VI only

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Item 42 refers to the following diagram which shows the position of some endocrine glands in humans.



42. In which of the following glands is follicle-stimulating hormone produced?

- (A) I
- (B) II
- (C) III
- (D) IV

43. Which of the following are true about the production of gametes in humans?

- I. Males produce more gametes than females.
- II. Females produce more gametes than males.
- III. Males produce gametes for a longer period of their life span than females.
- IV. Females produce gametes for a longer period of their life span than males.

- (A) I and III only
- (B) I and IV only
- (C) II and III only
- (D) II and IV only

44. During pregnancy, the developing foetus receives from its mother protection against infections, a perfectly balanced diet, warmth and oxygen. Which of the following can a mother no longer provide for her new-born infant?

- (A) Food
- (B) Oxygen
- (C) Warmth
- (D) Antibodies

45. Oral contraceptives prevent pregnancy by

- (A) destroying sperms
- (B) preventing ovulation
- (C) destroying released eggs
- (D) preventing implantation

46. You have found a flower which has a long style with a feathery stigma, long filaments, no corolla and very small sepals. This flower is adapted for

- I. cross-pollination
- II. self-pollination
- III. pollination by the wind
- IV. easy access by insects

- (A) I only
- (B) II only
- (C) I and III only
- (D) I and IV only

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47. Seeds from the same pod were planted under the same conditions. The plants which grew from them appeared to be very similar. What is the MOST likely explanation for this occurrence?

- (A) The seeds had been formed asexually after fertilisation.
- (B) The plant from which the seeds came became vegetatively propagated.
- (C) The same pollen grain had been responsible for fertilisation of the seeds.
- (D) The plant may have been self-pollinated.

Item 48 refers to the following information.

Some fowls (called Erminette) have black and white feathers evenly distributed over the body. A breeder crosses two of these fowls and finds that there are pure white, pure black and Erminette among the offspring.

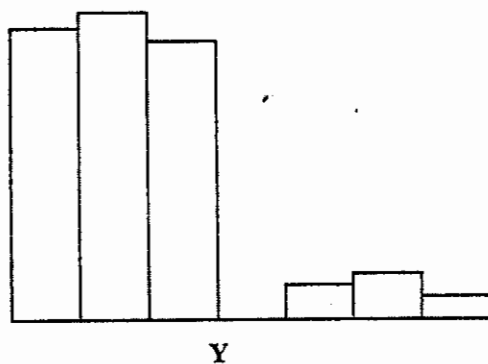
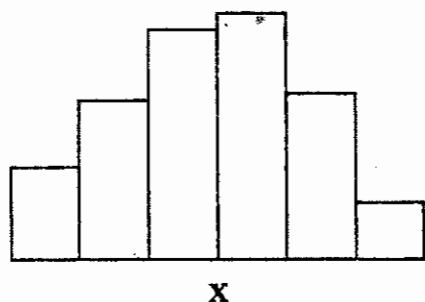
48. What would be the expected ratio of black:white:Erminette?

- (A) 1 : 1 : 1
- (B) 1 : 1 : 2
- (C) 1 : 2 : 1
- (D) 3 : 1 : 1

49. In which of the following parts of the plant does the process of mitosis occur?

- (A) Phloem
- (B) Xylem
- (C) The region behind the root hair
- (D) The region behind the root tip

Item 50 refers to the following diagrams, plotted from data collected on a survey of human characteristics.



50. X and Y can both be used to show different kinds of

- (A) variation
- (B) selection
- (C) inheritance
- (D) reproduction

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51. To which of the following groups does a disease caused by a pathogen belong?
- (A) Infectious
 - (B) Deficiency
 - (C) Hereditary
 - (D) Physiological
52. Which of the following is NOT one of the body's natural ways of defending itself against infection?
- (A) Clotting
 - (B) Vaccination
 - (C) Action of phagocytes
 - (D) Production of antibodies
53. Which of the methods listed below could help to reduce the spread of diseases caused by insect vectors?
- I. Sterilization
 - II. Use of insecticides
 - III. Use of disinfectants
- (A) I only
 - (B) II only
 - (C) I and II only
 - (D) I, II and III
54. Which of the following are **direct** consequences of disease in man?
- I. Loss of man hours and productivity
 - II. Loss of human life
 - III. Loss of livestock and crops
- (A) I and II only
 - (B) I and III only
 - (C) II and III only
 - (D) I, II and III
55. The following statements were made by a number of drug users.
- I. I drink strong coffee every hour. The caffeine keeps me going.
 - II. I have to inject myself with insulin twice a day.
 - III. I know alcohol is not good for me so I have only three drinks a day.
 - IV. I take antibiotics regularly in case I catch a cold.
- Which of these individuals could be considered drug abusers?
- (A) I and III only
 - (B) I, II and III only
 - (C) I, III and IV only
 - (D) II, III and IV only
56. Which of the following situations BEST exemplifies an ecological community?
- (A) A colony of termites
 - (B) A flock of migrating birds
 - (C) Flies on a piece of rotting meat
 - (D) The organisms in a freshwater pond
57. Which of the following statements is true of inorganic fertilisers?
- (A) They help to hold water in the soil.
 - (B) They improve the crumb structure of soil.
 - (C) They make nutrients available more quickly.
 - (D) They do not affect soil pH.

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58. Replanting trees on a bare hillside is an example of

- (A) conservation and afforestation
- (B) preservation and conservation
- (C) restoration and preservation
- (D) reafforestation and restoration

Item 59 refers to the following table.

Species	Number of Throws of a Quadrat									
	1	2	3	4	5	6	7	8	9	10
P	7	7	14	28	2	1	0	0	4	3
Q	18	21	36	3	10	3	1	5	5	6
R	10	18	46	0	12	10	8	2	16	4
S	1	0	0	0	1	3	4	1	21	2

59. The area in which the quadrat on the third throw fell is MOST probably

- (A) under a large shade tree
- (B) close to a walkway
- (C) a very sandy area
- (D) far from a source of water

60. What would be the BEST method of investigating the range of organisms found between the low tide mark and the high tide mark on a sloping, rocky sea shore?

- (A) Using a line transect
- (B) Using large quadrats
- (C) Taking a complete census of the area
- (D) Collecting as many species as possible

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.